



KungFlow

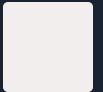
A Notification Firewall for Cognitive Overload

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VIDEO

The Problem

Constant interruptions destroy focus and increase stress.



Notifications and constant digital communication
increase stress and reduce uninterrupted focus time



Context switching and frequent app/tab switching
increase cognitive load

Why This Matters

Research-backed evidence

Interruptions disrupt focus and require significant recovery time

(PMC 2023)

Behavioral signals like mouse movement and typing patterns reliably indicate cognitive load

(HCI International 2018, ACM 2013, ScienceDirect 2009)

Reducing interruptions improves focus and performance

(Our problem validation experiment 2026)

Switching behavior increases cognitive load (ResearchGate 2010)

KungFlow: Adaptive Notification Firewall

Unlike manual Do Not Disturb, KungFlow reacts to your actual cognitive state in real-time.

GREEN STATE

User is not overloaded

Notifications pass through

YELLOW STATE

System is learning/calibrating

Rising load detected

RED STATE

User is overloaded

Firewall activates protection

Detecting Cognitive Load

Three behavioral signals that reveal your cognitive state.



MOUSE MOVEMENT

Speed, acceleration, micro-pauses
indicate stress



TYPING BEHAVIOR

Keystroke dynamics, backspace
frequency, error rates



APP SWITCHING

Frequency and patterns of switching
between different applications

No typed content is stored – only behavioral metadata.

The KungFlow Prototype

A working MVP with a clean, intuitive dashboard interface.



STATUS DASHBOARD

Shows green/yellow/red notification firewall state in real-time.



STATISTICS

Displays current activity-window metrics and behavioral data.



SETTINGS

User controls for data collection toggle and firewall options (Default Do Not Disturb, WhatsApp, Outlook).



PRIVACY PAGE

Explains behavioral metadata collection and data handling.

PROTOTYPE SCREENS

System Architecture

How KungFlow works under the hood.

01

Desktop App / Local Agent

A desktop app that collects behavioral metadata, responsible for notification firewall action and display status of the user.

02

Node.js / Express Server

Authenticates users, calculates cognitive-load score, updates state.

03

SQL Server DB

Stores users, sessions, metrics, cognitive state.

Why KungFlow Stands Out

Existing Solutions

- Manual Do Not Disturb: Requires manual activation, no behavioral sensing
- Freedom / Opal: Reactive and rule-based, not adaptive
- Focus Bear: Requires pre-defined schedules, no real-time detection

KungFlow's Advantage

- Adaptive, user-state-based detection
- Personalized baseline for each user
- Activates only when needed
- Lightweight behavioral metadata (no expensive biometrics)
- Real-time, automatic protection

KungFlow helps users protect focus at the exact moments they need it most.